INCH-POUND
MIL-K-43359E
INT. AMENDMENT 2(SH)
23 MARCH 1990
USED IN LIEU OF
AMENDMENT 1
1 APRIL 1988

MILITARY SPECIFICATION
KETTLES, STEAM-JACKETED (STAINLESS STEEL)
ELECTRICALLY HEATED

This interim amendment is approved for use within the Department of the Navy, Naval Sea Systems Command with MIL-K-43359E, dated 16 December 1986.

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2.1.1, under "Standards/Military", add the following:

"MIL-STD-167/1 - Mechanical Vibration of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements For the Control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of"

2.2, under "American Society of Testing and Materials": add the following:

"F 1166 - Standard Practice for Human Engineering Design for Marine Systems, Equipment and Facilities."

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3.6.5: Delete and substitute:

"3.6.5 Water level indicator. Each kettle shall be equipped with a device to indicate when the water in the steam-jacket is at a low level. Low level shall be the point at which there is still sufficient water in the jacket to cover the heating elements. The water level indicator shall be located in a position to be visible to the attendant. If a sight glass is used to indicate water level, it must have a permanent indeible marking to ensure the attendant can actually determine if a low water condition exists in the steam-jacket."



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3.8, User's Instructions (5), at the beginning add the following:

"Complete a daily check of the water level in the steam kettle prior to use. Fill reservoir if sight glass water level is low or if the low water indicator light is on."



Add the following paragraphs:

"3.14 Naval shipboard use. When the electrically heated steam-jacketed kettles are to be for shipboard use (see 6.2), the following paragraphs apply:

3.14.1 Electromagnetic compatibility. When specified for shipboard use (see 6.2), the units shall be designed and equipped for electromagnetic compatibility in accordance with the requirements of MIL-STD-461, class A4 for surface ships and class A5 for submarines. The equipment shall meet the emission and susceptibility requirements for CEO1, CEO3, and REO2 (see 4.5.4)

3.14.2 Inclined operation. The electrical kettles shall operate satisfactorily when inclined at an angle of 15 degrees ( ) each side of the vertical in each of two vertical planes at right angles to each other, with no spillage of fluid or product when tested (see 4.5.5).

3.14.3 Environmental suitability. When specified for shipboard use (see 6.2), electrically heated steam kettles shall withstand ships vibration and motion. Controls, switches, moving parts and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness, or damage when tested (see 4.5.6).

3.15 Human factors criteria. When shipboard use is specified (see 3.14 and 6.2), human factors engineering criteria, principles, and practices, as defined in ASTM F 1166, shall be used in the design of electrical kettles. Maintenance and operational activities shall permit safe and efficient performance by the 5th percentile female to the 95th percentile male as defined in sections 5.6 and 5.10 of ASTM F 1166. Sufficient clearance or free area required around an item shall permit an individual with applicable body dimensions and physical capabilities to safely operate, maintain, remove, or replace that item. When establishing accessibility requirements, both physical and visual access must be provided along with access for the use of any tools, test equipment or repair parts needed. Controls, switches, valves, and indicator lights needed for operation or maintenance shall be selected and integrated into the machines so as to meet the applicable requirements of sections 5.2 and 5.4 of ASTM F 1166. When inspecting for defects and performing tests (see section 4), the equipment shall adhere to human factors engineering considerations listed herein."



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Add the following paragraphs:

"4.5.4 Electromagnetic inference test. When required for shipboard use, the unit shall be tested by the contractor in accordance with test methods CE01, CE03 and RE02 of MIL-STD-462. The contractor shall furnish written certification that the equipment meets the requirements of MIL-STD-462. Non-conformance with the requirements specified shall constitute failure of the test.

4.5.5 Inclined operational test. Position the electric kettle with the base set at an angle of 15 degrees, then operate electric kettle for 30 seconds at each side of the vertical in each of two vertical planes at right angles to each other. At each of these positions observe for conformance with specified requirements in 3.14.2. For the test, the electric kettle shall be filled with water.

4.5.6 Shipboard environmental test. The electric kettle under normal operating conditions, shall be tested in accordance with MIL-STD-167/1, Type I equipment. The electric kettle shall be secured to the test machine in the same manner that it will be secured on shipboard (see 3.6.2.4). Failure of the electric kettle to perform its function during and after testing, or meeting requirements of 3.14.3, shall constitute failure of this test."

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5.1: Delete and substitute:

"5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2)."

5.1.1.1, Line 1: Delete "type I" and substitute "type II".
Line 4: Delete "type II" and substitute "type I".

5.1.2, Line 1: Delete "level C" and substitute "Commercial"

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5.2, Line 1: Delete "C" and substitute "commercial"

5.2.2, Delete "packing"

5.2.3, Line 1: Delete "Level C" and substitute "Commercial",
Delete "package as specified in 5.1."



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6.2, (o): Delete and substitute:

"o. Level of preservation and packing (see 5.1 and 5.2)."

6.2, (p): Delete.

6.2, add the following:

"q. When electric steam kettle is for shipboard use (see 3.14 d 3.15).
r. When electromagnetic compatibility is specified for shipboard use (see 3.14.1).
s. When environmental suitability is specified for shipboard use (see 3.14.3)."

Add the following paragraph:

"6.3.1 Technical manuals. When required for shipboard use, technical manuals shall be furnished for all units and shall be in accordance with the Technical Manual Contract Requirements for Food Service Equipment. At procurement, the contracting officer shall require the contractor to submit a copy of the proposed technical manual to the Naval Ship Systems Engineering Station (NAVSSES), Code 074E, Philadelphia, PA 19112-5083, for technical and format acceptance. After approval, two final manuals shall be packed with each unit and copies distributed per direction of the distribution list provided. If the technical manual has previously been approved by NAVSSES 074E and assigned a NAVSHIPS/NAVSEA number, and the equipment is identical as before, two copies only need be supplied with each to the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099. Minor approved changes to the technical manual will be made to existing technical manuals by the U.S. Navy and to all contracted and future manuals by the contractor. No equipment will be shipped without two copies of the approved technical manuals."

Preparing activity:
Navy - SH
(Project 7310-N783)



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